

Jiyeon Lee

(+82) 10-4117-0003 | termrad6170@gmail.com |

EDUCATION

Seoul National University

Seoul, Korea

B.S. in Computer Science — Major GPA: 4.30/4.30, Overall: 4.14/4.30

Mar. 2025 – Present

- **Presidential Science Scholar (National Merit):** Selected as 1 of 60 students nationwide for the highest-level undergraduate scholarship in Korea; awarded for exceptional research potential in Computer Science.
- **Relevant Coursework:** Discrete Mathematics, Data Structures, Linear Algebra, Principles of Accounting, Database, Big Data, Object-Oriented Programming, Introduction to Information Security and Blockchain.

PROJECTS

Gut Microbiome Health Classifier via CNN on Gram-Stained Images | *Tensorflow*

2024

- Induced gut microbiome variation in silkworm larvae using DSS at multiple concentrations and collected fecal microscopy images via Gram staining to build a labeled dataset.
- Designed and trained a multi-class CNN model with manual hyperparameter tuning, achieving **96% classification accuracy** in predicting larval health states from staining patterns alone.
- Validated model predictions against SEM-EDS elemental analysis (C, N, P, S ratios) and metagenomic sequencing (species diversity indices), confirming biological consistency.

Frame-Drop Strategy for Real-Time Autonomous Driving Event Captioning | *PyTorch*

2026

- Built a video captioning pipeline using ViT for frame-level visual embedding and a GPT-2-based transformer decoder to generate natural-language descriptions of autonomous driving scenes (BDD dataset).
- Systematically evaluated inference speed vs. quality tradeoff across 1, 2, 4, 8, and 16 fps sampling rates; found that 4 fps maintained average BLEU score of 0.534 while reducing inference time by **over 80%** (1.55ms → 0.28ms).
- Demonstrated practical viability of frame-dropping as a lightweight optimization for real-time deployment without retraining.

Virtual Wardrobe Web Application | *Django, Python, JavaScript*

2026

- Designed and deployed a full-stack wardrobe management web app enabling users to browse owned clothing from anywhere to encourage mindful consumption.
- Implemented secure user authentication with email verification, custom user model to resolve session conflicts, and hashtag-based search; iterated on schema and architecture mid-development based on usability findings.
- Conducted beta testing with real users, collected feedback, and shipped targeted fixes; deployed via web hosting.

AWARDS & HONORS

Presidential Science Scholarship for Undergraduate (Computer Science)

2025

Selected as 1 of 60 students nationwide in Korea for academic excellence and research potential in CS.

Kim Jae-chul AI Class, Seoul National University

2025

Selected among the top 30 students university-wide for an advanced AI curriculum program at SNU.

Silver Award, Korea Science & Engineering Fair International (KSEF)

2022

Awarded for research in biological classification using deep learning; competed at national level.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, HTML/CSS

Frameworks: Django, PyTorch, TensorFlow

Developer Tools: VS Code, Visual Studio, PyCharm, IntelliJ, CLion

Libraries: pandas, NumPy, Matplotlib